

# Haotian Xu 许皓天

*RL algorithms and RL infrastructure for long-horizon agents — sole owner of the RL framework, RL training, and credit-assignment algorithms behind Apodex 1.0 and Apodex 1.1.*

haotian@apodex.com   htxu91@gmail.com   yyht.github.io/personal-page   2,000+ citations · h-index 16 · i10 22

## SUMMARY

RL researcher and infrastructure engineer working on reinforcement learning for agents that run for thousands of steps. I built the asynchronous training stack for both Apodex flagship releases end to end, designed the fragment-level and token-level credit-assignment algorithms (PIVOT-RL, SAO), and ran the RL that produced the 1.0 → 1.1 benchmark jump — on a 397B-class MoE flagship and a 35B-A3B open-weight Mini. Co-author of OpenRLHF and 10+ papers at NeurIPS, ICLR, ACL, EMNLP, and TPAMI.

## RESULTS APODEX 1.0 → 1.1

The RL stage I owned is what separates the two releases. Against Apodex 1.0, the 1.1 flagship gains **+35.0 on FrontierScience-Research** (28.3 → 63.3), **+22.0 on APEX-Agents** (16.5 → 38.5, 2.3×), **+19.5 on GDPVal** (59.3 → 78.8) and **+14.0 on FrontierFinance** (40.3 → 54.3), with simulated-business net worth on YC-Bench rising **21.6×** (\$47,966 → \$1,038,255); HLE (49.0 → 56.1) and DeepSearchQA (84.6 → 92.4) moved up too, confirming the agentic gains cost nothing in general reasoning or search. The 35B-A3B open-weight Mini reproduces the pattern on the same recipe at a twelfth of the scale (FrontierScience-Research +26.7, FrontierFinance +17.0, APEX-Agents +12.3), which is the evidence that the recipe rather than the parameter count is carrying the result. Artificial Analysis independently ranks Apodex 1.1 at **Intelligence Index 44, #11 of 177 models** against a median of 18, with GDPval-AA v2 Elo 1348 — ahead of DeepSeek V4 Pro and Qwen3.7 Max.

## TECHNICAL CONTRIBUTIONS

### LAYER · INFRASTRUCTURE

#### Fully-asynchronous agentic RL framework

Designed and built the training framework for both releases on top of miles, against the defining pathology of agentic RL: trajectory wall-clock cost is wildly irregular across search, code execution, file processing, and multi-agent coordination, so any synchronous pipeline idles on its slowest episode.

- **Fully decoupled loops** — training workers and rollout engines run on independent loops, communicating only through a shared data buffer and an output queue. Completed trajectories enter optimization asynchronously, with no barrier on stragglers.
- **Non-blocking evaluation pool** — a dedicated pool scores checkpoints without ever interrupting the training loop, taking evaluation off the critical path.
- **Hybrid-colocate GPU reclamation** — idle training cards are recruited as generation engines, converting bubble time into rollout throughput.

### LAYER · CREDIT ASSIGNMENT · FRAGMENT

#### PIVOT-RL

The headline RL algorithm of Apodex 1.1. A long agentic rollout terminates in a single 0/1 verifier reward: terminal outcomes give only coarse supervision, and useful intermediate work routinely precedes a late failure. PIVOT-RL makes the

**trajectory fragment** the unit of credit — neither the whole trajectory nor the token.

- **Pivot discovery from self-rollouts**, not SFT demonstrations — locate the consequential decision points where the policy commits to an unproductive strategy.
- **Variance-based pivot screening** — sample multiple local continuations from each candidate state and use the mean and variance of their verified binary rewards to select states worth training on. Low-variance groups collapse group-normalized advantages to zero and yield no usable gradient.
- **Localized continuation tasks** — preserve the useful prefix, roll out and optimize only the continuation, with a short corrective hint that supplies directional guidance only: never a prediction target, and absent at inference time.
- **Stateful-environment restoration** — restore executable environment state at each pivot, so continuations are physically valid rather than merely plausible text.
- **Mixed curriculum** — interleave localized continuations with unhinted full-task problems, combining efficient learning at failure-relevant states with autonomous end-to-end problem solving.

LAYER • CREDIT ASSIGNMENT • TOKEN

## SAO and value pretraining

SAO (Async Single-rollout PPO) — the finer-grained companion layer, giving per-token advantage inside a fragment.

- Value model with GAE and **observation tokens skipped during credit propagation**, so advantage never flows through environment tokens the policy did not generate.
- **Large-scale value pretraining** to eliminate critic cold-start, the dominant failure mode of PPO on long agentic trajectories.
- **Two critic updates per policy update**, keeping the critic tracking a shifting policy distribution under asynchrony.
- **Frozen attention with MoE projections updated** — gradient stability and implicit regularization for the value head on MoE backbones.

LAYER • RELIABILITY

## Diagnostics for multi-week RL runs

Owned the instrumentation that kept 397B RL runs honest: rollout-structure collapse detection — for instance parallel multi-query degradation under penalty-only rewards combined with length-normalized objectives — plus reward-hacking audits, benchmark-leakage closure for SWE-bench Verified-class evaluations, and FP8-versus-BF16 inference parity checks on rollout engines.

---

### EXPERIENCE

**Apodex** *RL Algorithms & RL-Infra, Agentic RL*

2025.11 — Present

Formerly MiroMind. Sole owner of the RL framework, RL training, and credit-assignment algorithms for Apodex 1.0 and Apodex 1.1. Built the fully-async training stack, PIVOT-RL and SAO credit assignment; drove the 1.0 → 1.1 gains above on both the 397B flagship and the 35B-A3B open-weight Mini.

**Xiaohongshu — HiLab** *Math Reasoning & Agentic RL*

2024 — 2025.8

RL for mathematical reasoning and tool-using agents; agentic RL scaling laws for spontaneous code execution (NeurIPS 2025).

**Douyin / ByteDance** *AIGC Safety*

2023

|             |                |                                                                                                                                                                                      |
|-------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OPEN SOURCE | OPENRLHF       | Co-author. Ray-based scalable, high-performance RLHF framework. EMNLP 2025, 371 citations.                                                                                           |
|             | MILES-VALUES   | Dedicated SAO (Async Single-rollout PPO) framework with large-scale value pretraining.                                                                                               |
|             | ASYNC_OPENRLHF | Fully-asynchronous RLHF pipeline extending OpenRLHF.                                                                                                                                 |
|             | AGENTHARNESS   | Public evaluation harness reproducing Apodex-1.0 ReAct results across BrowseComp, BrowseComp-ZH, xbench-DeepResearch, HLE, SuperChem, FrontierScience, DeepSearchQA, and WideSearch. |

|                       |                                                                                              |                           |
|-----------------------|----------------------------------------------------------------------------------------------|---------------------------|
| SELECTED PUBLICATIONS | Apodex 1.1: Scaling Agentic Intelligence for Complex Work                                    | arXiv:2608.23283 · 2026   |
|                       | Apodex-1.0: A Verification-Centric Agent Team for Discoverative Intelligence                 | Apodex Tech Report · 2026 |
|                       | From System 1 to System 2: A Survey of Reasoning Large Language Models 529 cites             | IEEE TPAMI 2025           |
|                       | OpenRLHF: A Ray-based Easy-to-use, Scalable and High-performance RLHF Framework 371 cites    | EMNLP 2025                |
|                       | Agentic RL Scaling Law: Spontaneous Code Execution for Mathematical Problem Solving 71 cites | NeurIPS 2025              |
|                       | IterResearch: Rethinking Long-Horizon Agents with Interaction Scaling 44 cites               | ICLR 2026                 |
|                       | Search Self-Play: Pushing the Frontier of Agent Capability 25 cites                          | ICLR 2026                 |
|                       | GEM: A Gym for Generalist LLMs 17 cites                                                      | ICLR 2026                 |
|                       | Omni-DPO: A Dual-Perspective Paradigm for Dynamic Preference Learning 9 cites                | ICLR 2026                 |
|                       | TL;DR: Too Long, Do Re-weighting for Efficient LLM Reasoning Compression 10 cites            | ACL 2026                  |
|                       | Shuttle Between Instructions and Parameters                                                  | ACL 2026                  |

|           |                                                                 |             |
|-----------|-----------------------------------------------------------------|-------------|
| EDUCATION | <b>Tsinghua University</b> Department of Electronic Engineering | 2013 – 2016 |
|           | <b>UESTC</b> B.Eng.                                             | 2009 – 2013 |

|                   |                   |                                                                                                                                                                                                                                                             |
|-------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TECHNICAL PROFILE | RL ALGORITHMS     | PPO/GRPO and group-normalized variants, GAE, value pretraining, off-policy importance correction and token-level clipping, fragment- and turn-level credit assignment, RLHF, MCTS, DPO variants, reward modeling and anti-reward-hacking.                   |
|                   | RL INFRASTRUCTURE | Fully-async rollout/training decoupling, hybrid-colocate GPU reclamation, shared-buffer pipelines, Megatron and torch-dist checkpointing and conversion, TP/CP/EP parallelism, vLLM and SGLang rollout engines, FP8 inference parity, Ray, OpenRLHF, miles. |
|                   | LANGUAGES         | Chinese (native), English (professional).                                                                                                                                                                                                                   |